FINANCIAL MANAGEMENT

Suggested Answers
November-December 2024

Answer to the Question# 1(a)(i):

No rationing

Year	**0**	**1**	**2**	**3**	**4**
Time	**t0**	**t1**	**t2**	**t3**	**t4**
Discount factor @ 15%	1.000	0.870	0.756	0.658	0.572
	Tk.'000	**Tk.'000**	**Tk.'000**	**Tk.'000**	**Tk.'000**
Project A	(15,000)	(4,350)	9,072	3,948	1,716
Project B	(20,000)	(8,700)	18,900	16,450	14,300
Project C	(17,500)	4,350	8,316	9,212	5,720
Project D	(25,000)	6,090	6,804	8,554	1,716
Project E	(16,000)	(4,350)	1,512	18,424	13,156

	NPV Tk.'000	Rank
Project A	(4,614)	5
Project B	20,950	1
Project C	10,098	3
Project D	(1,836)	4
Project E	12,742	2

Therefore, accepting all projects with a positive NPV, accept projects B, C and E

Answer to the Question# 1(a)(ii):

Single-period capital rationing

Project	A	B	C	D	E
NPV (Tk.'000)	(4,614)	20,950	10,098	(1,836)	12,742
Investment t 0 (Tk.'000)	15,000	20,000	17,500	25,000	16,000
NPV/Tk.	-	1.05	0.58	-	0.80
Rank		1st	3rd		2nd

Therefore, project B can be accepted and the rest amount can be invested in Project E

Answer to the Question# 1(a)(iii):

Single-period capital rationing - inflows and outflows, negative NPVs

Using benefit cost ratios: NPV / Initial investment

	Benefit / cost	NPV per Tk. Invested	Ranking
Project A			*
Project B	Tk. 20,950 / 10,000	2.10	2
Project C			*
Project D	Tk. (1,836) / 7,000	(0.26)	**
Project E	Tk. 12,742 / 5,000	2.55	1

- Project A would never be accepted because it has a negative NPV and uses up funds in the restricted year. Project C would always be accepted since it has a positive NPV and releases funds in the restricted year. A total of Tk. 5,000,000 is then available.
- Project D has a negative NPV but releases funds at t1.
  Project D is accepted, this makes an extra Tk. 7 million available t1. However, in doing so a negative NPV (Tk. 1,836,000) is incurred. Thus, it is necessary to examine whether the extra positive NPV generated by the additional investment finance outweighs this cost.

(1) Available capital = Tk. 2,000,000. Accept Project C, E and 20% of B
Total NPV = Tk. 27,030,000
(2) If D is accepted the available capital becomes Tk. 14,000,000 [Tk. 2,000,000 + Tk. 5,000,000 (from project C)+Tk. 7,000,000 (from project D)]

Accept project C, D, E and 90% of B. Total NPV = Tk. 29,859,000

Answer to the Question# 1(a)(iv):

Indivisible projects

Possible investment portfolios are:
Project B or C or E (C and E)
The portfolio which has the highest NPV is C and E requiring an investment of Tk. 33.5 million and generating Tk. 22.8 million.

Answer to the Question# 1(b)(i):

Expected Portfolio Return Using CAPM

1. Calculate the Expected Return for Each Project Using CAPM:

CAPM Formula:
Expected Return = Risk-Free Rate + β × (Market Return - Risk-Free Rate)
For Project X:
Expected Return (X) = 7% + 1.7 × (11% - 7%)
= 7% + 1.7 × 4%
= 7% + 6.8%
= 13.8%

For Project Y:
Expected Return (Y) = 7% + 1.4 × (11% - 7%)
= 7% + 1.4 × 4%
= 7% + 5.6%
= 12.6%

2. Calculate the Portfolio Return:

Portfolio Return = (Weight of X × Return of X) + (Weight of Y × Return of Y)
- Given weights: 55% for Project X and 45% for Project Y
Portfolio Return = (0.55 × 13.8%) + (0.45 × 12.6%)
= 7.59% + 5.67%
= 13.26%
Conclusion: The expected return of the portfolio is 13.26%.

Answer to the Question# 1(b)(ii):

Applications of CAPM in Finance Theory

1. **Asset Pricing and Valuation**: CAPM is widely used to determine the fair value of an asset by relating its expected return to its risk. By applying CAPM, investors can identify if an asset is undervalued or overvalued based on its expected return and risk profile compared to the market.
2. **Portfolio Optimization**: CAPM helps in constructing an optimal portfolio that maximizes return for a given level of risk. By diversifying investments across assets with different betas, investors can achieve a balanced portfolio that mitigates unsystematic risk and aligns with their risk tolerance.

3. **Performance Evaluation**: CAPM provides a benchmark for evaluating the performance of investment managers. By comparing the actual returns of a portfolio to the expected returns derived from CAPM, investors can assess whether the manager has delivered excess returns (alpha) or underperformed relative to the risk taken.

Answer to the Question# 2(a):

Total dividends paid and the dividend per share under current dividend pay-out ratio:

Net income = BDT 700,000,000
Dividend payout ratio = 35%
Total dividends paid = Net income X Dividend payout ratio
= BDT 700,000,000 X 0.35
= BDT 245,000,000

Total shares outstanding = 25,000,000
Dividend Per Share (DPS) = Total dividends paid / Total shares outstanding
= BDT 245,000,000 / 25,000,000
= BDT 9.80

Payout ratio and the growth strategy and financial stability:

The current payout ratio of 35% aligns with the company's conservative approach, allowing it to retain a significant portion of earnings (65%) for reinvestment and expansion. This strategy ensures financial stability and supports long-term growth, especially crucial in a competitive industry like pharmaceuticals. By maintaining this balance, the company can navigate market uncertainties while still providing returns to shareholders.

Answer to the Question# 2(b):

Total dividends paid and the dividend per share under 50% dividend pay-out ratio:

Dividend payout ratio = 50%
Total dividends paid = Net income X Dividend payout ratio
= BDT 700,000,000 X 0.50
= BDT 350,000,000

Total shares outstanding = 25,000,000
Dividend Per Share (DPS) = Total dividends paid / Total shares outstanding
= BDT 350,000,000 / 25,000,000
= BDT 14.00

Impact of this increased payout ratio:

Increasing the payout ratio to 50% will reduce retained earnings, leaving the company with less internal funding for the expansion project. The company may need to seek additional external financing, potentially increasing its leverage and financial risk. This higher payout ratio could be attractive to shareholders seeking immediate returns but may affect long-term growth if not managed properly..

Answer to the Question# 2(c):

Number of shares to be issued under stock dividend:

Stock dividend = 20%
Total shares outstanding = 25,000,000
New shares issued = Total shares outstanding X Stock dividend
= 25,000,000 X 0.20
= 5,000,000

New total shares outstanding:

Previous outstanding shares + new issues
= 25,000,000 + 5,000,000
= 30,000,000

Adjusted market price per share:

Market price per share = BDT 120
Current market capitalization = New total shares outstanding X Market price per share
= 25,000,000 X 120
= BDT 3,000,000,000

Adjusted market price per share = Market capitalization / New total shares outstanding
= BDT 3,000,000,000 / 30,000,000
= BDT 100

Effect of stock dividend in the share price and investor perception:

The stock dividend increases the number of shares outstanding, diluting the share price. However, it rewards shareholders by increasing their shareholding without cash outflow, preserving funds for reinvestment. This may positively affect investor perception as the company demonstrates a commitment to growth while still providing value to shareholders.

Answer to the Question# 2(d):

Ethical Considerations of Changing Dividend Policy

1. **Balancing Shareholders' Interests and Long-Term Goals**: The board must consider how changes to the dividend policy align with the company's strategic objectives. Increasing the payout ratio may satisfy shareholders in the short term, but could hinder long-term growth if the company cannot reinvest sufficient funds.
2. **Employee and Community Well-being**: The company must ensure that financial decisions do not adversely impact employees' job security or the broader community. Reinvesting retained earnings can promote sustainable growth, job creation, and community development, aligning with broader social responsibilities.
3. **Financial Stability and Increased Leverage**: Taking on additional debt to finance expansion carries risks, including potential financial instability. The board must assess the impact of increased leverage on the company's balance sheet and ensure that the company can meet its debt obligations without compromising financial health.

Navigating these ethical considerations is crucial for maintaining trust and fostering long-term success. The board must weigh the potential benefits and risks of any changes to ensure decisions are in the best interest of all stakeholders.

Answer to the Question# 3(a):

Potential interest payments without hedging:

Q1: 50,000,000 X 0.052 X 3/12 = GBP 650,000
Q2: 50,000,000 X 0.055 X 3/12 = GBP 687,500
Q3: 50,000,000 X 0.057 X 3/12 = GBP 712,500
Q4: 50,000,000 X 0.060 X 3/12 = GBP 750,000

Total = 650,000 + 687,500 + 712,500 + 750,000
= GBP 2,800,000

Interest payments using proposed FRAs:

Q1 (No FRA): 50,000,000 X 0.052 X 3/12 = GBP 650,000
Q2 (3x6 FRA): 50,000,000 X 0.054 X 3/12 = GBP 675,000

Q3 (6x9 FRA): 50,000,000 X 0.058 X 3/12 = GBP 725,000
Q4 (9x12 FRA): 50,000,000 X 0.061 X 3/12 = GBP 762,500

Total = 650,000 + 675,000 + 725,000 + 762,500
= GBP 2,812,500

Comparison and evaluation:

Without hedging: GBP 2,800,000
With FRAs: GBP 2,812,500
Additional cost: GBP 12,500

The hedging strategy results in a slight additional cost of GBP 12,500. This is due to the conservative FRA rates chosen, which provide stability but are slightly higher than the expected LIBOR rates. Despite the additional cost, the strategy offers predictability and protection against potential higher rate increases.

Answer to the Question# 3(b):

Interest payments without hedging:

Q1: 50,000,000 X 0.050 X 3/12 = GBP 625,000
Q2: 50,000,000 X 0.058 X 3/12 = GBP 725,000
Q3: 50,000,000 X 0.060 X 3/12 = GBP 750,000
Q4: 50,000,000 X 0.063 X 3/12 = GBP 787,500

Total = 625,000 + 725,000 + 750,000 + 787,500
= GBP 2,887,500

With FRAs (Actual LIBOR Rates):
Interest payments using FRA rates remain the same as in part (a).

Comparison: Without hedging: 2,887,500 GBP With FRAs: 2,812,500 GBP
Net Financial Impact: Without hedging: 2,887,500 GBP With FRAs: 2,812,500 GBP Net Savings: 75,000 GBP
Hedging with FRAs provides savings of 75,000 GBP, demonstrating its effectiveness in mitigating interest rate risk under this scenario.

Answer to the Question# 3(c):

Strategic Benefits and Potential Risks of Using FRAs
Strategic Benefits:

1. Interest Rate Risk Management: FRAs provide a hedge against unfavorable interest rate movements, offering predictability and stability in interest payments.
2. Cash Flow Stability: By locking in interest rates, FRAs help maintain stable cash flows, aiding in financial planning and budgeting.
3. Cost Savings: Effective use of FRAs can result in cost savings if actual interest rates exceed the FRA rates.

Potential Risks:

1. Opportunity Cost: If actual interest rates are lower than the FRA rates, the company may incur higher costs than without hedging.
2. Counterparty Risk: As over-the-counter instruments, FRAs carry the risk of counterparty default.
3. Complexity and Costs: Structuring and managing FRAs require expertise and may involve transaction costs.

Comparison with Interest Rate Swaps:
FRAs:

- Short-term hedging instruments.
- Flexible and tailored to specific periods.
- Easier to implement for short-term exposures.

Interest Rate Swaps:

- Longer-term hedging instruments.
- Fixed-floating interest rate exchanges for longer durations.
- More suitable for long-term debt instruments and exposure management.

Recommendations:

- Use FRAs for short-term hedging to manage quarterly interest rate exposures.

- Consider interest rate swaps for long-term debt and exposure to achieve comprehensive risk management.
- Regularly review and adjust the hedging strategy based on market conditions and financial objectives.

This comprehensive approach allows London Textile Limited to balance risk management, financial stability, and growth objectives effectively.

Answer to the Question# 4(a)(i):

The bank is being asked to buy the Korean won and will give the exporter:
= 15,000,000/12.25
= Tk. 1,224,490 in exchange

Answer to the Question# 4(a)(ii):

The bank is being asked to sell the Indonesian Rupiah to the importer and will charge for the currency:
= 100,000,000/131.18
= Tk. 762,311 in exchange

Answer to the Question# 4(b)(i):

The receipt using a forward contract is fixed with certainty at 20,000,000 / 8.30 = Tk. 2,409,639
This applies to both exchange rate scenarios.

Answer to the Question# 4(b)(ii):

The cost of the option is Tk. 24,000. This must be paid at the start of the contract.
The results under the two scenarios are as follows:

Scenario	(a)	(b)
Amount received at exchange rate	20 million @ Peso 9/Tk. = Tk. 2,222,222	@ Peso 7.60 / Tk. = 2,631,579
Amount received at exercise price	20 million @ Peso 8.28/Tk. = Tk. 2,415,459	@ Peso 8.28 / Tk. = 2,415,459
Does the company exercise the option	Yes	No

	(a) Tk.	(b) Tk.
Taka received	2,415,459	2,631,579
Less: option premium	(24,000)	(24,000)
Net receipt	2,391,459	2,607,579

Answer to the Question# 4(b)(iii):

The results of not hedging under the two scenarios are as follows:

Scenario	(a)	(b)
Exchange rate	9.00	7.60
Taka received	2,222,222	2,631,579

Answer to the Question# 4(c):

The option premium makes foreign currency option quite expensive, and could therefore be unsuitable for any company trading on narrow profit margins. Options are often used by companies faced with:

(i) a currency exposure that might not arise at all or
(ii) where the amount of the total receipt or payment in uncertain.

Option premium = 210 x 1.2% = 2.52 VND
Worst case = 210 - 2.52 = 207.48 VND

If the spot rate in six-months' time is 215, the company will allow the option to lapse, and buy VND at the spot rate.

Answer to the Question# 5(a):

Calculation of NPV:

Particulars	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Initial investment	-120,000					
Resale value						
Working capital investment	-15,000					
Compliance cost	-10,000					
Government grant						
Revenue		90,000	151,410	132,300	134,820	134,820
Operating Costs		-45,000	-65,325	-65,325	-50,250	-50,250
Maintenance Costs		-5,000	-4,000	-4,000	-4,000	-4,000
Foregone investment return		-6,000	-6,000	-6,000	-6,000	-6,000
Opportunity Revenue		10,000	10,000	10,000	10,000	10,000
Upgrade costs			-2,000		-3,000	
Marketing campaigns		-3,000				
Contingency costs		-1,600	-1,600	-1,600	-1,600	-1,600
Depreciation		-20,000	-20,000	-20,000	-20,000	-20,000
Taxable Income		19,400	62,485	45,375	59,970	62,970
Tax 15% (Y1-3) 25% (Y 4-5)		2,910	9,373	6,806	14,993	15,743
EBIT less tax		16,490	53,112	38,569	44,978	47,228
Add depreciation	-145,000	36,490	73,112	58,569	64,978	67,228
Discount Factor @ 14%	1	0.8772	0.7695	0.6750	0.5921	0.5194
Present value	-145,000	32,009	56,258	39,532	38,472	34,916
Net Present Value	56,186					
Revenue		**Y1**	**Y2**	**Y3**	**Y4**	**Y5**
Base figure		90,000	140,000	140,000	120,000	120,000
Add Inflation of 5%			147,000	147,000	126,000	126,000
Add/ (Less) economic factor			3%	-10%	7%	7%
Adjusted revenue		90,000	151,410	132,300	134,820	134,820
Operating Cost		45,000	65,000	65,000	50,000	50,000
Assume 50% split for raw material and labour						
Projected raw material cost		22,500	32,500	32,500	25,000	25,000
Projected labour cost		22,500	32,500	32,500	25,000	25,000
Increased raw material cost (1.03)		22,500	33,475	33,475	25,750	25,750
Decreased labour cost (0.98)		22,500	31,850	31,850	24,500	24,500
Projected operating cost		45,000	65,325	65,325	50,250	50,250

Notes:

1. The initial feasibility cost of BDT 5 million is considered a sunk cost and is therefore excluded from the cash flows.
2. The cost of debt is already integrated into the Weighted Average Cost of Capital (WACC), and as such, it is not considered separately.
3. The reported revenue and operating costs have been adjusted for inflation.
4. Since the investment and returns are in the local currency, the depreciation of the local currency against foreign currencies has no effect on the cash flows.
5. All figures are presented in thousands of BDT (BDT '000).

Importance of NPV in capital budgeting:

Advantages:

NPV accounts for the time value of money, ensuring that future cash flows are valued at their present worth. It also considers all cash flows over the project's life, providing a complete financial picture. Furthermore, a positive NPV indicates that the project is expected to generate more value than its cost, making it a sound investment.

Limitations:

NPV relies on accurate estimates of future cash flows and discount rates. Inaccuracies can significantly impact the results. It does not explicitly account for project risk or uncertainty, though these can be factored into the discount rate to some extent. Moreover, the choice of discount rate is critical and can greatly affect the NPV calculation.

Role of NPV in investment decisions:

NPV is crucial in investment decisions for several reasons: It helps determine whether a project adds value to the company by comparing the present value of cash inflows to the initial investment. NPV allows for easy comparison of multiple projects, enabling the selection of the most financially viable option. As a quantitative measure, NPV provides an objective criterion for investment decisions, minimizing biases.

Overall, NPV is a valuable tool in capital budgeting, providing a clear indicator of a project's potential financial benefits. It helps companies make informed investment decisions by assessing the value addition and financial viability of projects.

Answer to the Question# 5(b):

Calculation of IRR using interpolation method:

Year	Cash Flow	PV at 20%	PV	PV at 30%	
	-		-		
0	145,000	1	145,000	1	-145,000
1	36,490	0.8333	30,408	0.7692	28,069
2	73,112	0.6944	50,772	0.5917	43,262
3	58,569	0.5787	33,894	0.4552	26,659
4	64,978	0.4823	31,336	0.3501	22,750
5	67,228	0.4019	27,017	0.2801	18,831
			28,428		-5,430
IRR	0.20	+	28,428	x	0.1000
			33,857		
			0.08		
	28.40%				

Comparison with required rate of return:

The project has an IRR of 28.40%, while the required rate of return is 14%. Since the IRR is above the company's required rate of return of 14%, it indicates that the project exceeds the company's minimum required return threshold, making it a viable investment opportunity.

The IRR method:

The IRR method calculates the discount rate at which the present value of future cash flows equals the initial investment, making the NPV zero. It is a common method for evaluating investment projects. IRR is used to assess the profitability of potential investments. A project with an IRR higher than the required rate of return is considered favorable. When comparing multiple projects, those with higher IRRs are generally preferred.

Strengths:

Time value of money: Like NPV, IRR accounts for the time value of money.
Simplicity: It provides a single percentage return, making it easy to compare with the required rate of return or other investment opportunities.
Intuitive decision-making: A higher IRR indicates a more profitable project, simplifying decision-making.

Weaknesses:

Multiple IRRs: Projects with non-conventional cash flows (e.g., alternating positive and negative cash flows) can have multiple IRRs, complicating the analysis.
Reinvestment assumption: IRR assumes that intermediate cash flows are reinvested at the same rate as the IRR, which may not be realistic.
Scale of investment: IRR does not consider the scale of investment, potentially favoring smaller projects with higher rates of return over larger, more profitable projects with lower rates of return.

Usefulness of IRR in financial decision-making:

IRR is useful in financial decision-making because it provides a clear, easily interpretable measure of a project's expected rate of return. It helps companies compare the profitability of various projects and make informed investment decisions. By comparing the IRR to the required rate of return, companies can assess whether a project meets their financial criteria and aligns with their strategic goals.

While IRR is a valuable tool in evaluating investment projects, it is essential to consider its limitations and use it alongside other financial metrics, such as NPV, to make well-rounded investment decisions.

Answer to the Question# 5(c):

Sensitivity analysis on NPV:

To assess the sensitivity of the project to changes in the discount rate, the discount rate will be varied by +/- 4% from the original 14%, and the impact on the NPV will be observed.

Year	Cash Flow	PV at 16%	PV	PV at 26%	
	-		-		
0	145,000	1	145,000	1	-145,000
1	36,490	0.8621	31,457	0.7937	28,960
2	73,112	0.7432	54,334	0.6299	46,052
3	58,569	0.6407	37,523	0.4999	29,279
4	64,978	0.5523	35,886	0.3968	25,780
5	67,228	0.4761	32,008	0.3149	21,169
			46,208		6,240
IRR	0.20	+	28,428	x	0.1000
			39,968		

0.07

27.11%

Project sensitivity to the changes:

Yes, the project is sensitive to changes in the discount rate. The NPV varies significantly with changes in the discount rate:

Purpose of sensitivity analysis:
Sensitivity analysis evaluates how changes in key assumptions or variables impact the project's outcome. It helps identify which variables significantly influence the project's financial performance, providing insights into the project's robustness under different scenarios.

Sensitivity analysis and its contribution to more informed investment decisions:
Risk assessment: Sensitivity analysis helps in understanding the project's risk by demonstrating how changes in key assumptions impact the NPV.

Robust decision-Making: It assists in making more informed decisions by highlighting the sensitivity of the project to changes in critical variables.

Scenario planning: Sensitivity analysis allows companies to prepare for different scenarios, improving preparedness for adverse conditions.

Confidence building: It increases confidence in investment decisions by providing a range of possible outcomes and their implications.

In summary, sensitivity analysis is a valuable tool for evaluating the potential variability in project outcomes. By understanding how sensitive the NPV is to changes in the discount rate, companies can make more informed and confident investment decisions.

Answer to the Question# 6(a):

Market capitalization

Cost of equity shares:
= (0.60 x 1.05) / 6.30 + 0.05
= 15%

	Tk. '000
Present value of future dividends without the project (i.e., market value)	6,300
Add: Present value of extra cash receipts used to increase dividends	
Year 1 Tk. 300,000 x 0.870	261
Year 2 Tk. 500,000 x 0.756	378
Year 3 Tk. 300,000 x 0.658	197
Present value of dividends with project accepted	7,136

Thus market capitalization of the company rises to Tk. 7,136,000

Answer to the Question# 6(b):

Shareholders' gain

	Tk. '000
Market capitalization with project	7,136
Market capitalization without project	(6,300)
	836
Less: cash subscribed for rights issue	(620)
Gain made by shareholders	216

Answer to the Question# 6(c):

Net present value of the project

Period	Cash Flow (Tk.'000)	DF @ 15%	PV (Tk.'000)
Year 0	-620	1.000	-620.00
Year 1	300	0.87	261.00
Year 2	500	0.756	378.00
Year 3	300	0.658	197.40
NPV			216.40

Therefore, gain made by shareholders is equal to NPV of the project.

Answer to the Question# 6(d):

Shareholder's gain, taking up rights

a. Value of the company after project accepted = Tk. 7,136,000
Number of shares in issue = 1.1 million
Price per share after right issue = Tk. 7,136,000 / 1.1 million
= Tk. 6.4873

Effect on holder of 100 shares

		Tk.
Value of shareholding after issue	110 x Tk. 6.4873	713.60
Value of shareholding before issue	100 x Tk. 6.30	(630.00)
Increase in shareholding value		83.60
Cash subscribed for issue	10 x Tk. 6.20	(62.00)
Gain made		21.60

b. Value of the company after project accepted = Tk. 7,136,000
Number of shares in issue = 1.2 million
Price per share after right issue = Tk. 7,136,000 / 1.2 million
= Tk. 5.9467

Effect on holder of 100 shares

		Tk.
Value of shareholding after issue	120 x Tk. 5.9467	713.60
Value of shareholding before issue	100 x Tk. 6.30	(630.00)
Increase in shareholding value		83.60
Cash subscribed for issue	20 x Tk. 3.10	(62.00)
Gain made		21.60

Answer to the Question# 6(e)(i):

Price of rights

The shareholder should hope to sell his rights for the difference between the rights price and the share's final value i.e., Tk. (5.9467 - 3.10) = Tk. 2.8467

Answer to the Question# 6(e)(ii):

Effect on holder of 100 shares

		Tk.
Value of shareholding after issue	100 x Tk. 5.9467	594.67
Value of shareholding before issue	100 x Tk. 6.30	(630.00)
Increase in shareholding value		(35.33)
Cash subscribed for issue	20 x Tk. 2.8467	56.93

Gain made	21.60

Answer to the Question# 6(f):

Minimum price of rights

If the shareholder is not to make a loss, he must recover Tk. 35.33 from the sale of the rights. The right should therefore be sold for:

Tk. 25.33 / 20 = Tk. 1.7665 each

---The End---